

Helmholtz Cage for CubeSat Magnetic Testing

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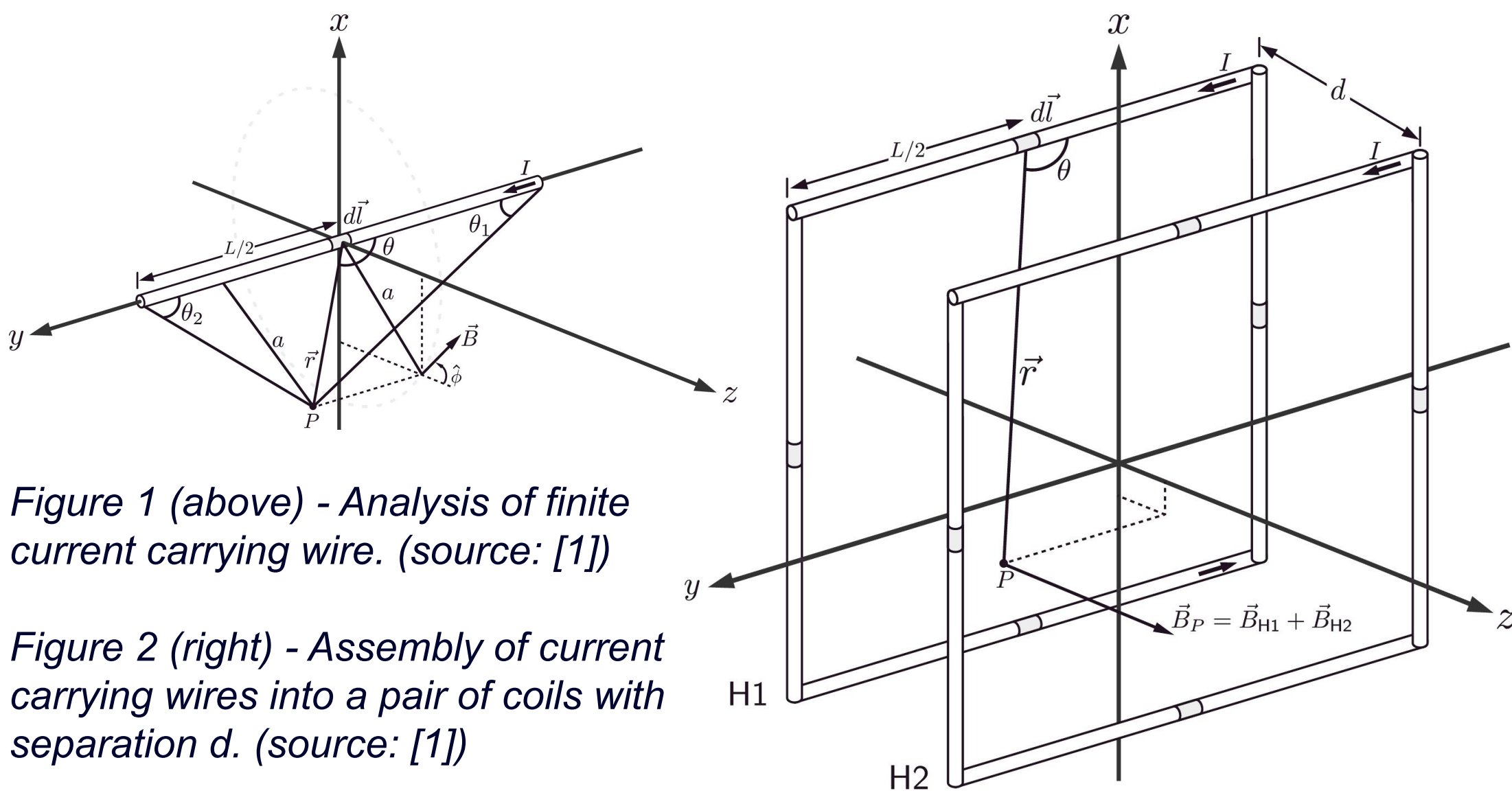
Introduction and Background

ALEASAT is a 1U CubeSat being developed by the UBC Orbit and SFU SAT design teams. Our mission is to provide Earth-observation imaging for disaster relief. Amateur radio operators will be able to request images directly from ALEASAT.

ALEASAT must undergo rigorous magnetic testing to ensure it meets design requirements and thus, requires a suitable magnetic testing apparatus. A Helmholtz Cage is a device that generates uniform magnetic fields, enabling a test environment that accurately reflects magnetic conditions in Low Earth Orbit. We are developing a Helmholtz Cage to ensure the efficacy of ALEASAT for long-term use.

Magnetic Field Model

It is required that the Device Under Test (DUT) fits within the magnetic field uniformity region produced by the Helmholtz cage. To ensure the Helmholtz cage can provide a sufficiently large uniformity region to support 1U, 2U, and 3U CubeSats, we developed a Python simulation to quantify the spatial variation of the magnetic field magnitude produced by a single pair of Helmholtz coils (see Figure 3). This simulation enables us to explore the impact of key design parameters (current, number of turns, coil size, coil spacing) on field uniformity and field strength.



Method:

The Biot-Savart law was used to determine the magnetic field produced by the finite current carrying wire in Figure 1. The resulting magnetic field is:

$$\vec{B} = -N \frac{\mu_0 I}{4\pi a} \hat{\phi} \int_{\theta_1}^{\theta_2} \sin \theta d\theta = N \frac{\mu_0 I}{4\pi a} (\cos \theta_2 + \cos \theta_1) \hat{\phi}$$

Each straight wire segment in Figure 2 will have its own magnetic field contribution. Using superposition, the straight wire segments can be combined into a pair of coils. With careful attention to orientation and geometry, superposition yields the total magnetic field at any point P in the Helmholtz coil pair setup.

Result:

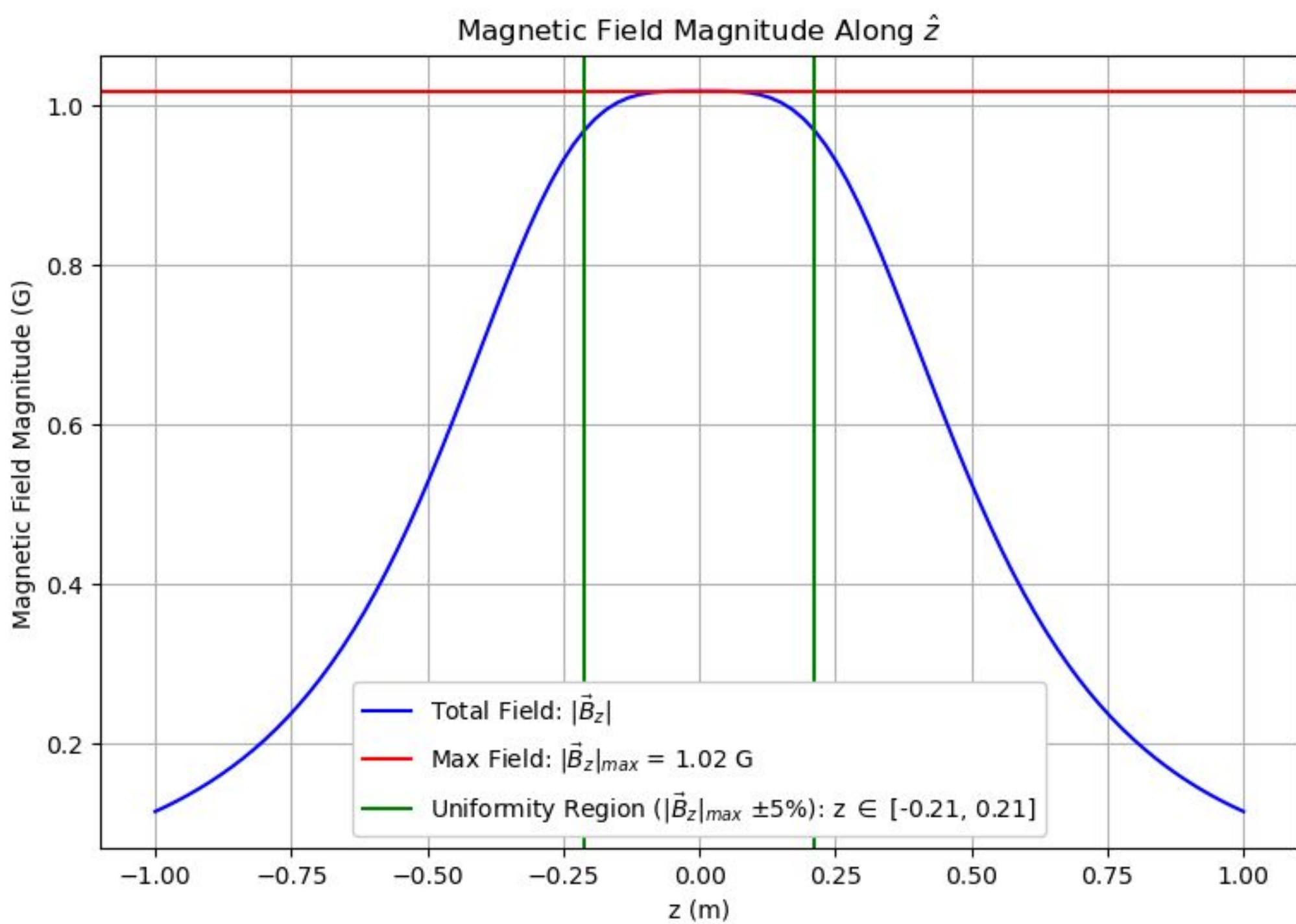


Figure 3 - Results of our single axis Helmholtz coil magnetic field simulation. Parameters: L = 80cm, d = 43.56cm, I = 1A, N = 50 turns.

The resulting square uniformity region for the example parameters in Figure 3 has a side length of 42.42cm.

Control and Feedback System

The control and feedback system is responsible for generating the precise desired magnetic field inside the Helmholtz cage. Since the magnetic field produced by each axis is directly proportional to the current through its coils, we control the system by regulating coil currents.

As shown in Figure 4, each axis of the cage is driven by a power supply, with each coil's current direction controlled by a corresponding H-bridge.

To provide feedback and ensure safe operation, a magnetometer is placed in the cage and thermocouples are mounted on the coils. The magnetometer is used for live feedback control of the produced magnetic field, while the thermocouples monitor coil temperatures to prevent overheating during operation.

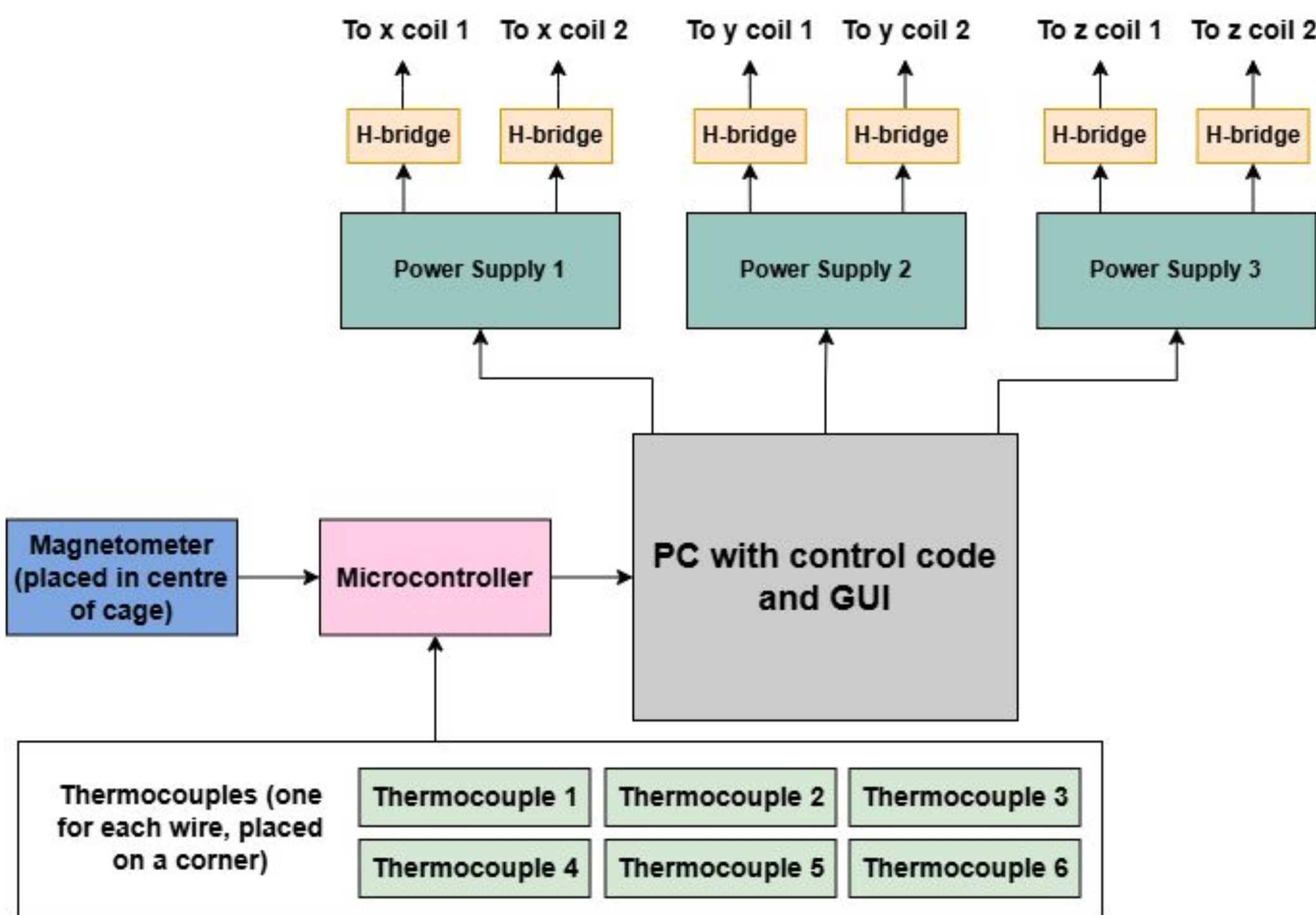


Figure 4 - Helmholtz cage system level design diagram for one of our design options.

Structure

A square Helmholtz cage was selected for the ease of fabrication and assembly. It also features improved structural stability compared to a circular design while maintaining a sufficiently large and uniform magnetic field region for CubeSat testing.

Apparatus Testing

To validate the Helmholtz cage design methodology, a single-axis prototype coil of 60 cm x 65 cm with 45 turns was constructed and tested experimentally.

Acknowledgement

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Method: Coil was driven by a DC power supply with ~3A current and magnetic field measurements obtained via a magnetometer. Field strength was measured at multiple positions within and around the coil to characterize the uniformity region. The field strength is approximately 1.6 G at the coil center.

Result: Experimental measurements of the prototype coil produced a field magnitude and uniformity profile consistent with simulation, confirming the validity of Helmholtz cage design.

We also plan to conduct noise characterization tests by driving the cage using Pulse Width Modulation to further validate our design.

Future Work/Goals

Future work will focus on construction and implementation of a full size 3 axis prototype with a magnetometer-feedback based control loop. We will implement a simple user interface for operating the cage and integrate an Earth magnetic field model for test campaigns. The testing framework will be expanded to support automated subsystem-level and integrated satellite testing. The completed Helmholtz cage will enable dipole moment characterization, magnetometer calibration, and magnetorquer validation, and will serve as a reusable magnetic testing platform for future CubeSat missions and research application beyond ALEASAT.

Reference / Bibliography

[1] Hurtado-Velasco, R., & Gonzalez-Llorente, J. (2016). Simulation of the magnetic field generated by square shape Helmholtz coils. Applied Mathematical Modelling, 40(23-24), 9835–9847. <https://doi.org/10.1016/j.apm.2016.06.027>



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